SVSM 2008 Mathematical Modeling

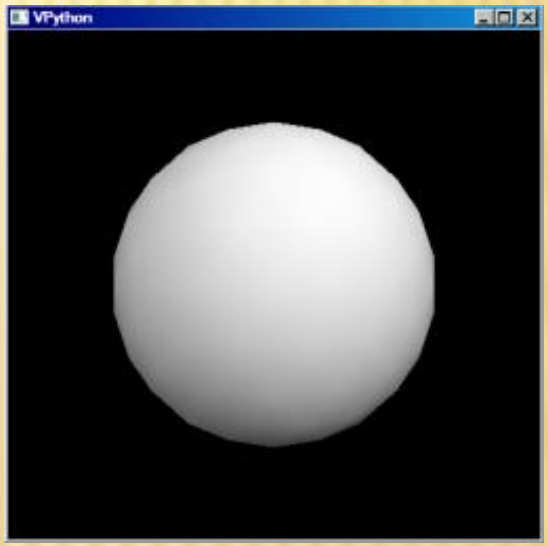
MODELING MOTION IN VPYTHON

VPYTHON = VISUAL PYTHON

- Easy to learn
- 3D Interactive modeling
- Python – background language
- Main site http://vpython.org/
- Download http://vpython.org/win_download25.html
- Python Site http://www.python.org/

GETTING STARTED

- Install Python and then VPython
- Run IDLE
- First line
  - from visual import *
- Next line
  - sphere()
- Save – **CTRL-S**
  - Save as **test.py**
- Run – **F5**



CHANGE SPHERE ATTRIBUTES

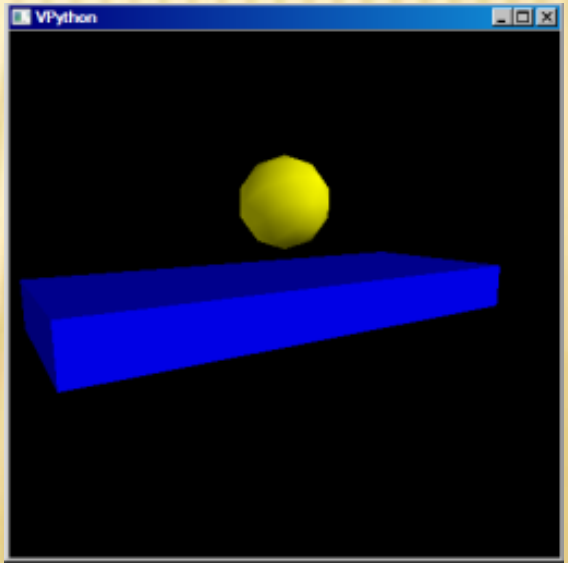
- Color
  - sphere(color=color.red)
- Radius
  - sphere(radius=0.5,color=color.red)
- Name
  - ball = sphere(radius=0.5,color=color.red)
- Position
  - ball = sphere(pos=(0,2,0),radius=0.5,color=color.red)
- Change position
  - ball.pos = (1,2,3)

NAVIGATION

- Zoom
  - Hold middle button and move mouse
  - Or, hold both left and right buttons
- Rotate
  - Hold right button

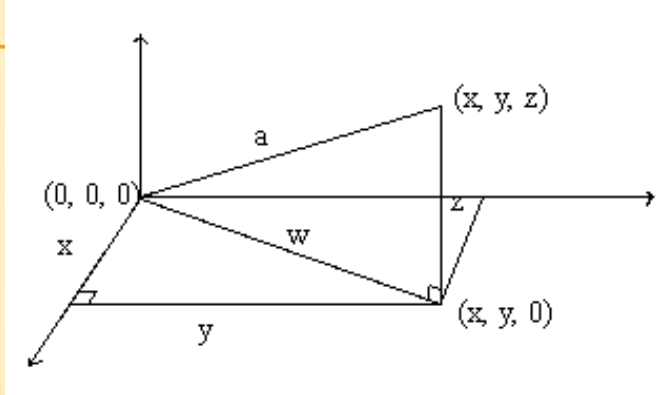
SIMPLE SCENE

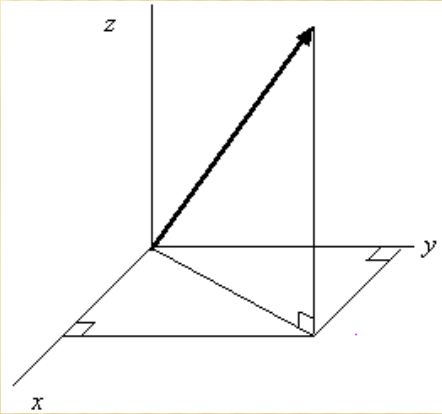
```
from visual import *

#Create a sphere
ball = sphere(pos=(0,2,0),color=color.yellow,radius=1)

#Create a box
floor = box(length=10, height=2, width=4,color=color.blue)
```

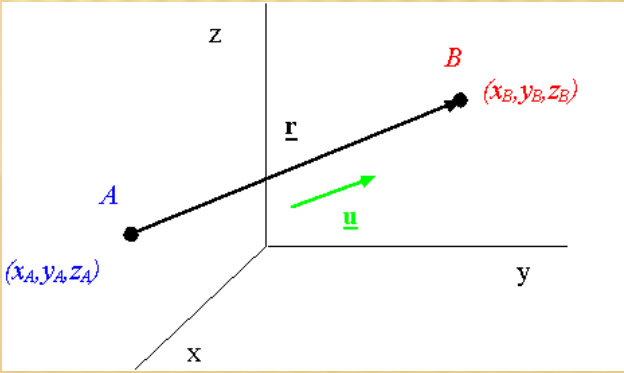


VECTORS

- Cartesian Coordinates
  - Locate points with (x,y,z)
- Vectors
  - Draw arrow from origin
  - Vectors have magnitude and direction
  - (1,2,3)
- General Vector
  - Draw from one point to another
- Examples
  - Position, velocity, force







VECTORS IN VPYTHON

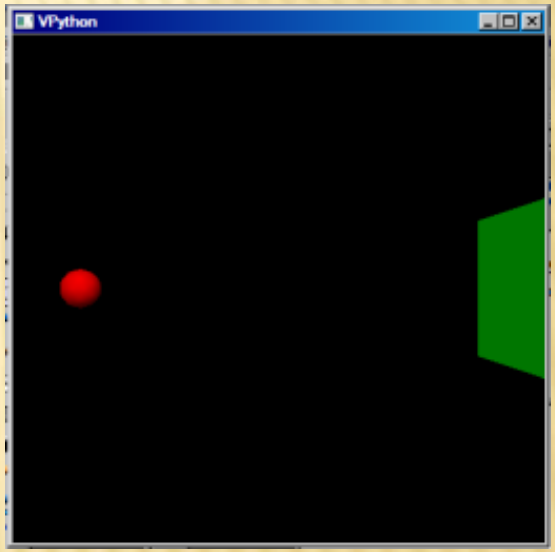
- Draw arrow from origin to center of ball
  - arrow(pos=(0,0,0), axis=ball.pos)
  - Begins at pos
  - Ends at axis
- Arbitrary arrow
  - arrow(pos=(1,2,3), axis=(0,2,-1))
- Arrow attributes
  - arrow(pos=(1,2,3), axis=(0,2,-1), shaftwidth=1,headwidth=2,headlength=3)

MOVING BALL – CONSTANT VELOCITY

ADDING MOTION

Goal: Make ball move and bounce off wall

- Create new program
  - File – New Window
  - Save as motion.py
- Create scene



motion.py - C:/Python25/Lib/site-packages/visual/examples/motion.py

File Edit Format Run Options Windows Help

```
from visual import *
ball = sphere(pos=(-5,0,0), radius=0.5, color=color.red)
wallR = box(pos=(6,0,0), size=(0.2,4,4), color=color.green)
```

CONSTANT VELOCITY

- Velocity
  - Velocity = displacement/elapsed time
  - $v = \Delta x/\Delta t$
- Rewrite
  - Δx=new position – old position
  - $\Delta x = v\Delta t$
  - new position = old position + $v\Delta t$
- In Vpython
  - **ball.pos = ball.pos + ball.velocity*dt**

MOVING BALL

- Need several time steps
  - Use while loop
    - Keep executing commands over and over
- Additional Information
  - Specify dt and velocity
  - Need rate – while loop executed 100 times/sec

motion.py - C:/Python25/Lib/site-packages/visual/examples/motion.py

File Edit Format Run Options Windows Help

```
from visual import *
ball = sphere(pos=(-5,0,0), radius=0.5, color=color.red)
wallR = box(pos=(6,0,0), size=(0.2,4,4), color=color.green)

dt = 0.5
ball.velocity = vector(.2,0,0)
while (1==1):
    rate(100)
    ball.pos = ball.pos + ball.velocity*dt
```

MAKING BALL BOUNCE

- Logical Tests
  - If the ball moves to far to right, reverse its direction.
- If statement
  - **If ball.x > wallR.x:**
  - Then what?
- Reverse direction
  - **ball.velocity = -ball.velocity**

```
while (1==1):
    rate(100)
    ball.pos = ball.pos + ball.velocity*dt
    if ball.x > wallR.x:
        ball.velocity.x = -ball.velocity.x
```

MODIFY PROGRAM

- Assignment 1
  - Add **wallL** on left at (-6,00)
  - Add test to have ball bounce off this wall too.
- Assignment 2
  - Change ball velocity to move ball at an angle with no *z*-component
- Assignment 3
  - Add a ceiling and a floor that touch vertical walls
  - Add back wall
  - Add invisible front wall (i.e., only use if statement)
  - Run program with ball bouncing off all walls.

UNIFORM MOTION – ADDING ACCELERATION

FREE FALL

Goal: Drop a ball that bounces on the floor.

- Free fall
  - All objects near the surface of the Earth fall with an acceleration -9.8 m/s^2, or -32 ft/s^2 (or a = -g).
- Acceleration
  - $a = \Delta v/\Delta t$
- Rewrite for VPython
  - new velocity = old velocity + $a\Delta t$
  - **ball.velocity = ball.velocity + ball.acceleration*dt**

BOUNCING BALL

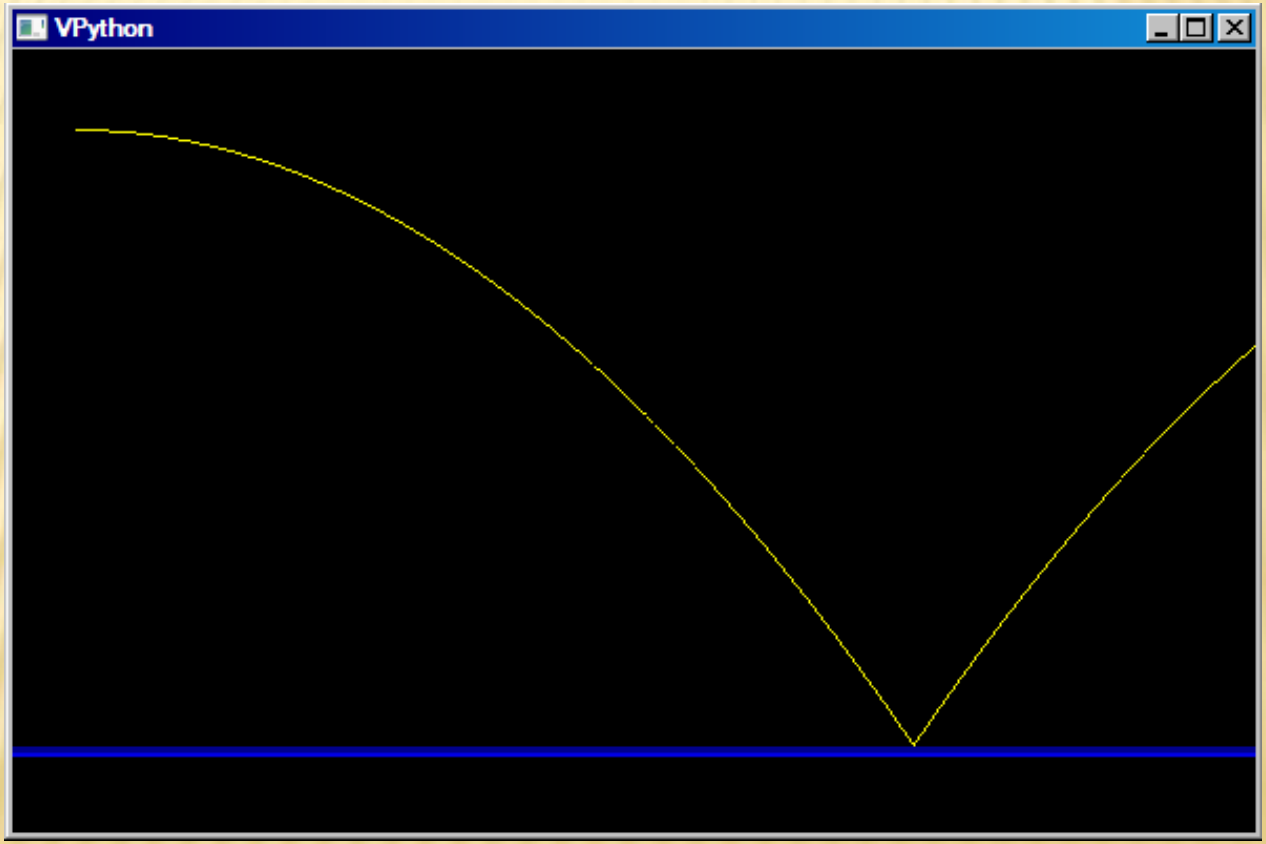
bounce.py - C:\Python25\Lib\site-packages\visual\examples\bounce.py

File Edit Format Run Options Windows Help

```
from visual import *
scene.width = 300
scene.height = 400
scene.autoscale = 0
scene.range = (100,100,100)
scene.center = (0,50,0)

floor = box(length=100, height=0.5, width=4, color=color.blue
ball = sphere(pos=(0,100,0),radius=2, color=color.red)
ball.velocity = vector(0,0,0)

dt = 0.01
while 1:
    rate(100)
    ball.pos = ball.pos + ball.velocity*dt
    if ball.y < 1:
        ball.velocity.y = -ball.velocity.y
    else:
        ball.velocity.y = ball.velocity.y - 9.8*dt
```

PROJECTILE MOTION

Free fall with constant horizontal velocity.



PROGRAM FOR PROJECTILE MOTION

bounce.py - C:\Python25\Lib\site-packages\visual\examples\bounce.py

File Edit Format Run Options Windows Help

```
from visual import *
scene.width = 600
scene.height = 400
scene.autoscale = 0
scene.range = (100,100,100)
scene.center = (0,50,0)

floor = box(length=300, height=0.5, width=4, color=color.blue)
ball = sphere(pos=(-90,100,0),radius=2, color=color.red)
ball.velocity = vector(30,0,0)
ball.trail = curve(color=color.yellow)

dt = 0.01
while 1:
    rate(100)
    ball.pos = ball.pos + ball.velocity*dt
    if ball.y < 1:
        ball.velocity.y = -ball.velocity.y
    else:
        ball.velocity.y = ball.velocity.y - 9.8*dt
    ball.trail.append(pos=ball.pos)
```

ERROR IN CODE?

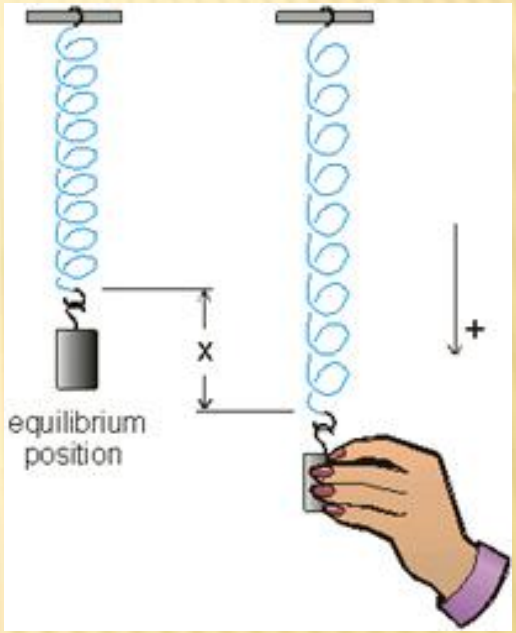
```
floor = box(length=300, height=0.5, width=4, color=color.blue)
ball = sphere(pos=(-90,100,0),radius=2, color=color.red)
ball.velocity = vector(30,0,0)
ball.trail = curve(color=color.yellow)

ball2 = sphere(pos=(-90,100,0),radius=2, color=color.red)
ball2.velocity = vector(30,0,0)
ball2.trail = curve(color=color.green)

ball3 = sphere(pos=(-90,100,0),radius=2, color=color.red)
v = vector(30,0,0)
ball3.trail = curve(color=color.blue)
a = vector(0,-9.8,0)
```

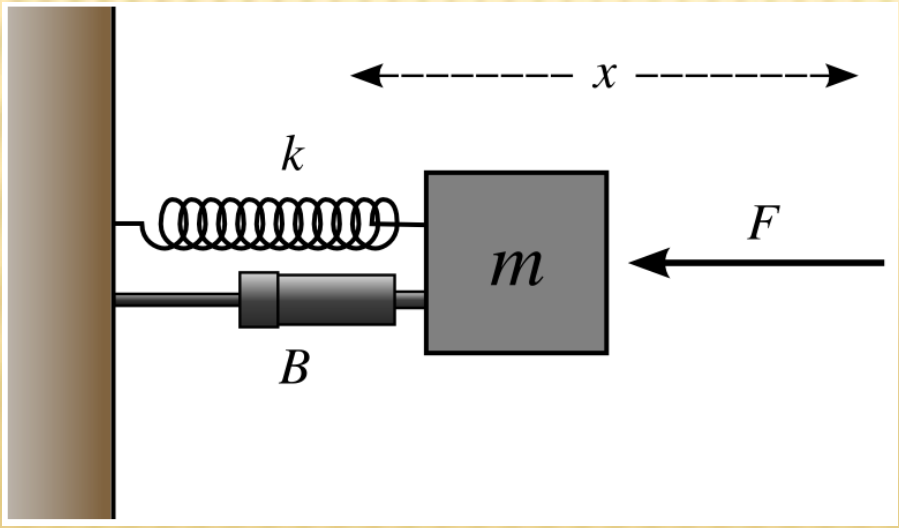
```
 dt = 0.01
while 1:
    rate(100)
    ball.pos = ball.pos + ball.velocity*dt
    if ball.y < 1:
        ball.velocity.y = -ball.velocity.y
    else:
        ball.velocity.y = ball.velocity.y - 9.8*dt
    ball.trail.append(pos=ball.pos)

ball.pos = ball.pos + ball.velocity*dt
 vel0=ball2.velocity
    if ball2.y < 1:
        ball2.velocity.y = -ball2.velocity.y
    else:
        ball2.velocity.y = ball2.velocity.y - 9.8*dt
    ball2.pos = ball2.pos + (vel0+ball2.velocity)*dt/2
    ball2.trail.append(pos=ball2.pos)

    if ball3.y < 1:
        v.y= -v.y
    else:
        v += a * dt
    ball3.pos += v* dt + .5 * a * dt**2
    ball3.trail.append(pos=ball3.pos)
```

MASS – SPRING SYSTEM

MASS-SPRING SYSTEM

- Hooke's Law for Springs
  - F = -kx
- Newton's 2nd Law of Motion
  - F=ma
  - Or, a= F/m





CREATE THE SCENE

```
from visual import *

scene.range = (1,1,1)
scene.center = (0,.1,0)
scene.width = 800
scene.height = 275
scene.background = (1,1,1)

x0 = vector(.65,0,0)
Surface = box(size=(2,.02,.5),pos=(0,-.1,0))
wall = box(size=(.04,.5,.3),pos=(-.77,.15,0))
spring = helix(pos=(-.75,0,0),axis=x0,
    radius=.08,coils=6,thickness=.01,color=color.blue)
block = box(pos=(0,0,0),size=(.2,.2,.2),color=color.red)
```

INITIALIZE

```
k = 10
m = 10
b=2
v = vector(0,0,0)
a = vector(0,0,0)
F = vector(0,0,0)

block.pos=(0.25,0,0)  # Set Initial position of block and connect spring
x = block.pos
spring.axis = x0 + x
```

ADD MOTION

```
finished = False
dt = .01
while not finished:
    rate(100)

    F = -k*x - b*v
    a = F/m
    v += a * dt
    x += v + .5 * a * dt**2

    block.pos = x
    spring.axis = x0 + x
```

CHANGE PROGRAM

- What happens when the mass is changed?
- What happens when *k* is changed?
- Change the initial block position
  - i.e., block.pos=(0.25,0,0)
  - Do you need to change other seetings?

MAKE UP YOUR OWN MOVING SYSTEM

FREE FOR ALL #1

Play with Vpython and create your own moving objects:

- Use walls, surfaces, and other objects
- Other built-in objects http://vpython.org/webdoc/visual/index.html

  Cylinder, arrow, cone, pyramid, sphere, ring, box, ellipsoid, helix

FREE FOR ALL #2

Seach for Vpython Code and see what it does at http://people.uncw.edu/hermanr/SVSM.htm